Fiscal Year:	FY 2016	**Task Last Updated:**	FY 05/12/2021
PI Name:	Furst, Eric Ph.D.		
Project Title:	Investigating the Structure of Paramagnetic Aggregates from Colloidal Ellipsoids-3 (InSPACE 3)		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	19716	**Congressional District:**	1
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	98-HEDS-03
Start Date:	03/01/2010	**End Date:**	04/19/2018
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:	McQuillen, John	**Contact Phone:**	216-433-2876
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Flight Program:	ISS		
Flight Assignment:	InSPACE ISS		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX16AD21G ; NNX10AE44G ; NNX07AD02G (InSPACE2)		
Performance Goal No.:			
Performance Goal Text:			

InSPACE is a microgravity fluid physics experiment that was performed on the International Space Station (ISS). InSPACE began in December 1997 and concluded with the third series of follow-on InSPACE experiments aboard the ISS in November 2013.
[Ed. note April 2021: added to Task Book when received information. Note that this investigator's (Dr. Eric Furst) original grant was InSPACE-2: Structural and Rheological Transitions of Field-Responsive Fluids in Microgravity, grant NNX07AD02G; period of performance was 1/8/2007-1/07/2010.

Project continued with Dr. Furst as PI 3/1/2010-1/29/2016 as Investigating the Structure of Paramagnetic Aggregates from Colloidal Ellipsoids-3 (InSPACE 3), grant NNX10AE44G; 10/20/2015-4/19/2018, grant NNX16AD21G with

Task Description:	same title, Investigating the Structure of Paramagnetic Aggregates from Colloidal Ellipsoids-3 (InSPACE 3)] InSPACE-2 was a continuation of the InSPACE investigation begun on ISS Expedition 6, providing new and improved samples for operation in the Microgravity Science Glovebox (MSG). The use of external fields to control the microstructure of colloidal suspensions has long been recognized as a powerful means for tailoring the mechanical, optical, and electronic properties of materials. Magnetorheological (MR) suspensions, in particular, provide a striking example. These normally stable fluids undergo a dynamic transition to a solid within milliseconds after the application of an external magnetic field. They are also important models for developing methods of bottom-up fabrication of micro- and nano-structured materials and devices using field-directed self-assembly. Investigating the Structure of Paramagnetic Aggregates from Colloidal Emulsions-2 (InSPACE-2) experiments focused on the structure of magnetically-polarizable particle suspensions over long times in steady (DC) and pulsed magnetic fields. In pulsed fields especially, the long-time kinetics of the suspension micro-structural coarsening provided new and important results. The InSPACE-2 experiments were the first to yield information on the full, three-dimensional aggregation process over timescales much longer than those that are accessible on the ground, which are limited by catastrophic sedimentation. Furthermore, the experiments identified a novel and intriguing dynamic instability in which the suspension microstructures were observed to buckle at specific field frequencies and field strengths. For InSPACE-3, we will investigate the effect of particle shape on the micro-structural evolution of MR suspensions. Recent ground-based experiments demonstrate a startling effect that particle shape has on the interactions of dipolar chains and the resulting suspension microstructure. Specifically, we will investigate suspensions of paramagnetic ellipsoid-shaped particles. In combination with the results of InSPACE-2, we hypothesize that particle shape will dramatically alter the aggregation kinetics, microstructures, and microscopic mechanics. This will potentially lead to the ability to engineer enhanced properties of these suspensions, including suppression of the lateral aggregation in magnetorheological fluid-based electromechanical devices or the ability to create new colloidal materials through field-directed self-assembly.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Space Applications: The quick phase-shifting property of magnetic colloidal fluids makes them potentially useful for devices such as optical interfaces, active noise, and vibration dampers inside spacecrafts. Future uses could include robotics, energy-transfer devices such as clutches, and other control systems. Earth Applications: The work has application to directed self-assembly of crystalline structures from particles that may eventually allow creation of new nano-materials fabricated from nanoparticle building blocks, with potential applications in medicine, energy storage, chemical separations, and catalysis. Magnetic colloidal fluid technology is currently being used for shock absorbers in racecars and sport cars. This technology is expanding to include making large-scale building foundation stabilizers for areas prone to earthquakes.
Task Progress:	[Note (May 2021): excerpted from PI's March 2016 Final Progress report for grant NNX10AE44G] The objective of InSPACE-3 is to study the effect of particle shape on the kinetics of aggregation and structures formed by pulsed magnetic fields in suspensions of super-paramagnetic particles. Based on the science results of InSPACE-2, the research of the InSPACE-3 project hypothesis is that particle shape will alter the aggregation kinetics, microstructures and microscopic mechanics, such as buckling instabilities that are observed for suspensions of polarizable spheres. New dynamics and structures formed by self-assembly using non-spherical colloidal building blocks could lead to enhanced properties of the colloidal aggregates, such as the ability to suppress lateral aggregation in magnetorheological fluid-based electromechanical devices or the ability to create new functional materials that control the flow of electromagnetic and thermal energy. During this project, ellipsoidal magnetic colloids were synthesized and samples sent to the International Space Station (ISS). Experiments were performed on ISS expeditions 32-40 that studied the suspension structure in toggled fields. A total of 52 experimental conditions were studied, spanning an experimental matrix that included ellipsoidal particle aspect ratio, field strength, and field frequency. In addition to these microgravity experiments, ground-based studies were performed, and results from earlier flight experiments (InSPACE-2) were analyzed and published. The InSPACE-3 project has been a success. The ISS experiments performed from 2012 through 2014 returned a rich set of results. These continue to be analyzed and will be reported in the archival literature. The project has also had a strong impact on Earth applications. Despite the effects of gravity in ground experiments, we showed that analogous two-step kinetics of phase separation we discovered in space station experiments were present for quasi-two-dimensional suspensions (magnetic suspensions in a gravitational field.) Our work led to a series of publications examining the out-of-equilibrium and emergent dynamics of these colloidal systems. From this, we have discovered important methods to guide the self-assembly of new functional materials, including new phononic crystals. Closely related theory and experiments, especially microscopic models of colloidal dynamics, impacted the measurement of colloidal interactions in kinetically arrested colloidal suspensions, biophysical experiments of molecular mechanics, and colloidal electrophoresis.
Bibliography Type:	Description: (Last Updated: 01/27/2026)
Articles in Peer-reviewed Journals	Swan JW, Vasquez PA, Furst EM. "Buckling instability of self-assembled colloidal columns." Phys Rev Lett. 2014 Sep 26;113(13):138301. https://doi.org/10.1103/PhysRevLett.113.138301 ; PMID: 25302919 , Sep-2014
Articles in Peer-reviewed Journals	Bauer JL, Liu Y, Kurian MJ, Swan JW, Furst EM. "Coarsening mechanics of a colloidal suspension in toggled fields." J Chem Phys. 2015 Aug 21;143(7):074901. https://doi.org/10.1063/1.4927563 ; PMID: 26298150 , Aug-2015
Articles in Peer-reviewed Journals	Shindel MM, Swan JW, Furst EM. "Calibration of an optical tweezer microrheometer by sequential impulse response." Rheol Acta. 2013 May;52:455-65. https://doi.org/10.1007/s00397-013-0698-2 , May-2013

Journal/Magazine covers	Bauer JL, Liu Y, Kurian MJ, Swan JW, Furst EM. "Cover in Journal of Chemical Physics for article, 'Coarsening mechanics of a colloidal suspension in toggled fields.' " J Chem Phys. 2015 Aug 21;143(7):074901. https://doi.org/10.1063/1.4927563 , Aug-2015